# DHRUV CHHABRA

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## Work Experience

**Quantitative Trader - Commodities;** Alpha Alternatives Holdings Pvt Ltd — May 2024 – Sept 2025

- **Multi-Asset Pairs Trading & Risk Framework**
  - Generated **₹20 lakh+ P&L** in 2025 on a ₹6 Cr book at a **1.9 Profit Factor** via early mean-reversion strategy
  - Protected the book from tail-risk losses via volatility-adjusted stop-losses, holding a **45% win rate** profitably
  - Partnered with discretionary traders to convert trade intuition into systematic **gradient-based momentum signals**
  - Delivered the project end-to-end after the intern's exit by auditing and rebuilding it into a systematic signal framework
- **Algorithmic Strategy Refinement and Portfolio Optimisation system**
  - Achieved **1.38% net monthly return** at a 1.2 Sharpe by rebalancing weights via Markowitz mean-variance optimisation
  - **Reduced operational errors and slippage cost ∼7x** (0.15% to 0.02%) by automating signal generation
  - Prevented deployment of flawed strategies by auditing a trend-following Futures algorithm, rectifying backtesting biases

**Associate Data Scientist;** Optum Global Solutions Ltd — Aug 2021 – May 2024

- **Automated Clinical Documentation & Validation System**
  - Cut clinical documentation time **by 55-60%** with a Gen AI engine automating medical claim-appeal summaries
  - Built a correctness-evaluation ML model to score AI summaries, **cutting manual audit effort** by 85%
  - **Scaled the engine** to cover 73% of all medical claim-appeal categories, widening the automation footprint
  - Validated **AI non-inferiority** to clinician summaries by designing a double-blind experiment for a 2-sample t-test
- **Prediction of Risk of Hospital Readmission**
  - Built a **$10-20M projected-savings case** for cutting avoidable 30-day readmissions via targeted patient intervention
  - **Doubled the hit rate** of intervention targeting (50% precision vs 23% base rate) via a readmission risk model
  - Presented the **cost-benefit analysis** and risk drivers to the division's research director, shaping the intervention strategy
- **Prediction of First Occurrence of Multiple Diseases Simultaneously**
  - Enabled early screening across 63 chronic diseases at once by designing a multi-label neural network on patient history
  - Demonstrated feasibility of single-model multi-disease prediction as a POC to the ML/AI R&D leadership

## Product & Strategy Projects

**smallcase: Strategy Performance - User Execution Gap** — Aug 2026

- Scoped a Zerodha-first MVP to separate strategy returns from users' actual performance, a gap validated by app reviews
- Attributed the performance gap to price, quantity and costs by reconciling user execution with strategy trades
- Added an LLM-powered Q&A layer for granular performance analysis, grounded in deterministic financial calculations

**Honasa Consumer: Turnaround and Margin Teardown** — July 2026

- Found the two-year rise in EBITDA margin from 7% to a record 11.3% came from cost lines, not from pricing
- Showed the recovery is volume-led: 30% volume growth vs 27.7% revenue in Q4 FY26, with realisation per unit falling
- Showed the FY31 margin target leans on cost efficiency and mix, raising doubts on the premiumisation narrative

**The Mental Health Coverage Gap in Indian Health Insurance** — May 2026

- Exposed a near-0% outpatient-therapy coverage gap in Indian retail health insurance by auditing best-in-class policies
- Showed outpatient mental-health cover can be economically viable via a B2B benefits platform redesign with hybrid funding

**Strategic Roadmap: Protecting Margins Amidst Global Silver Shortages** — Apr 2026

- Traced structural EBITDA risk for Solar PV manufacturers to a cumulative silver deficit of 700+ Moz built since 2021
- Found cost pass-through structurally fails beyond $118/oz silver by stress-testing margins across forward price scenarios
- Recommended a zero-cost collar, miner offtake deals, and silver-reduction R&D to fix costs and secure supply

**Algorithmic Trading with Backtesting Analysis** — Jan 2024

- Built and back-tested systematic trading strategies (alphas) using technical indicators to generate buy/sell signals
- Back-tested the results against S&P500 as the benchmark and computed the draw-downs and risk-adjusted returns

## Skills

`Python`, `SQL`, `C++`, `PySpark`, `Git`, LaTeX, LLM/Gen AI tooling, Statistical Testing, Advanced Excel (VBA)

## Education

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| **MSc, Statistics** | Indian Institute of Technology Kanpur | `8.33 /10` | July 2019 - May 2021 |
| **BSc, Statistics (Hons.)** | PGDAV College, University of Delhi | `9.47 /10` | July 2016 - May 2019 |

## Key Courses

Financial Statistics  Statistical Methods for Business Analytics  Statistical Simulation & Inference  Econometrics

## Extra Curricular

- Awarded the **Junior Diploma** in Hindustani Classical Music **(Violin)** from Prayag Sangeet Samiti, 2025
- Competed in multiple Ultimate Frisbee club nationals and **represented Maharashtra** at the **Bharat Trophy**, 2025